

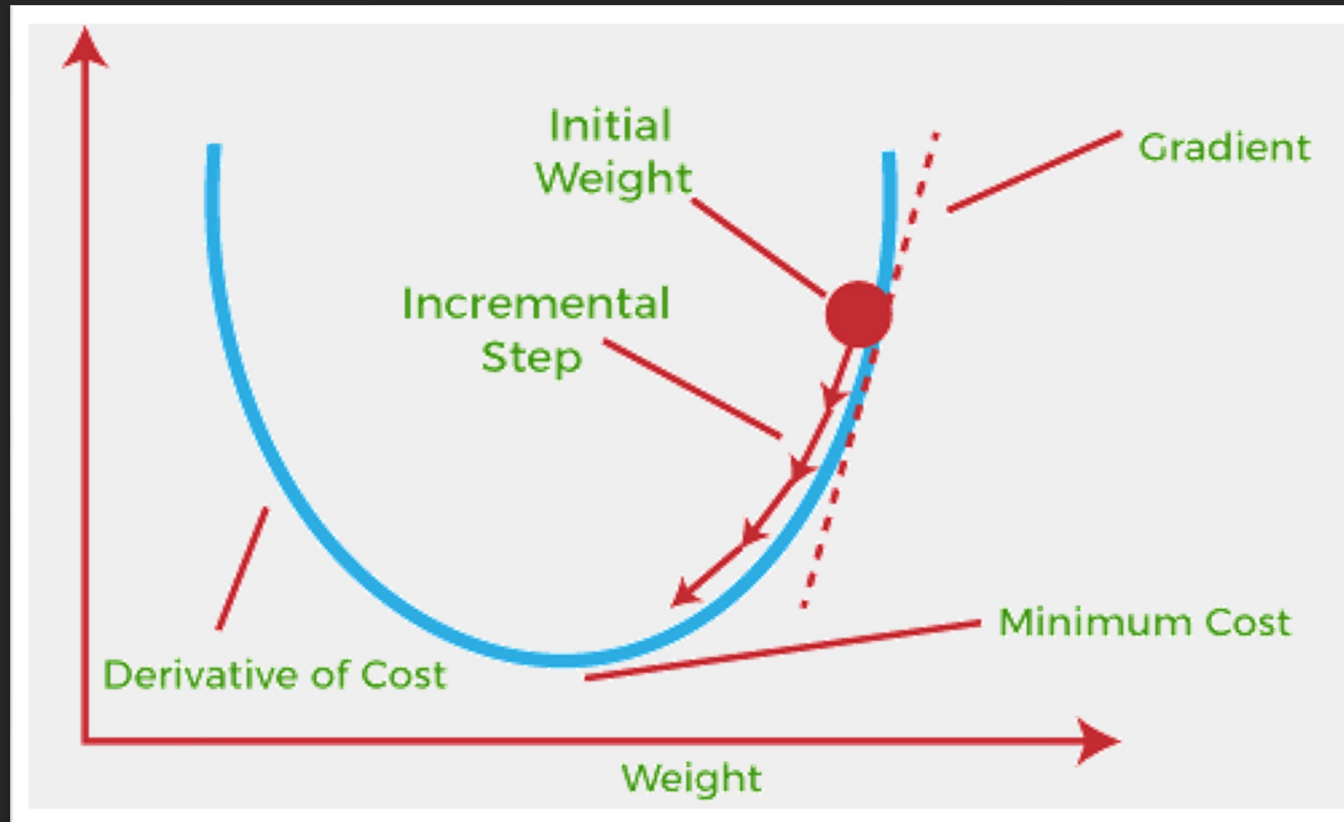
COM3021 – Data Science at Scale

Batch and Online Machine Learning

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Gradient descent



Batch learning

- Neural networks are trained over a series of epochs.

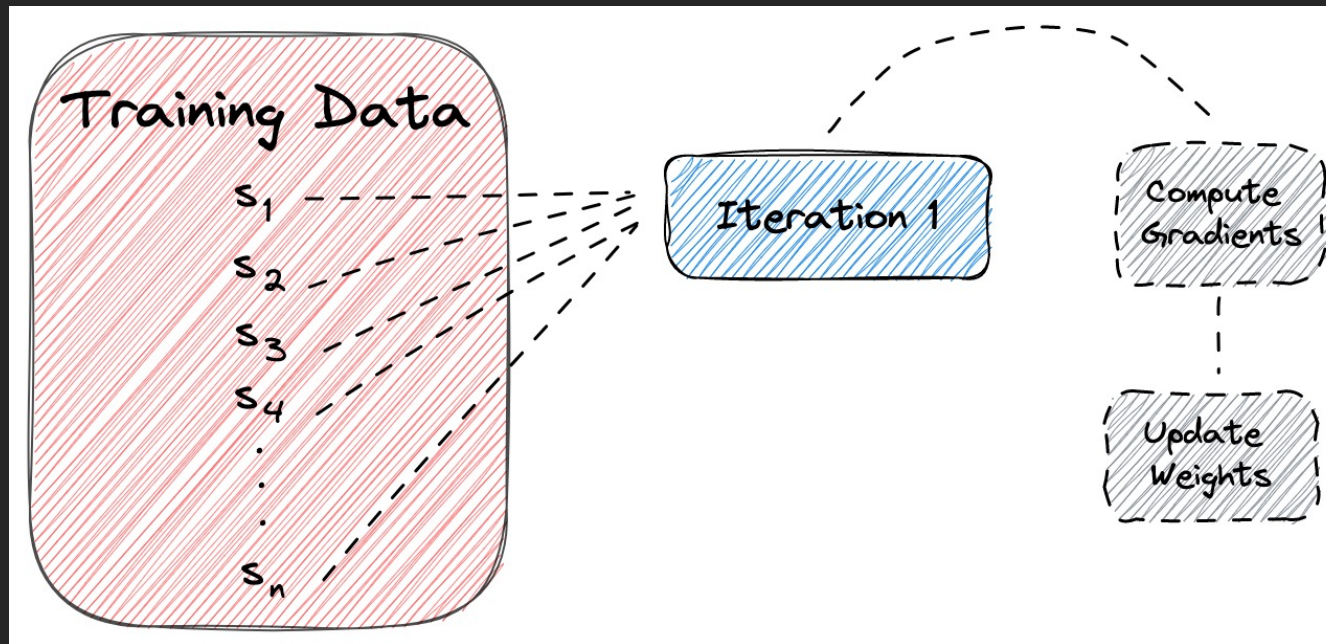
Each epoch will involve one forward pass and one backpropagation pass over the training data

In each epoch we can either

- *Compute the **true gradient** using all the available data*
 - **Full batch learning**
 - *Compute an **approximation of the gradient***
 - **Mini-batch learning**
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Full batch learning

- Uses all the data
- Computes the true gradient



Full batch learning

- Training machine learning models frequently operate through *batch learning*

All the available data is used

Training before the deployment

1. *Simpler to reason about*
 2. *Smoother and more consistent convergence*
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Full batch learning

- However, the data (or the problem) might not be static

Recommendation system: consumers preferences change over time

- *Concept Drift*

- We need to update a ML model due to these changes

- Cons

If we always use all the available data for training, retraining will become increasingly costly

Data will have to fit in memory

Mini-batch

- Alternatively, we can use more recent data for training (mini-batch)

It will be less computationally intensive

The older data won't affect the model as much

- Incremental learning
- How often should a model be retrained?
Daily, weekly, monthly?
- How much data should we use?
Week, month, year?

- Cons

More complexity added in

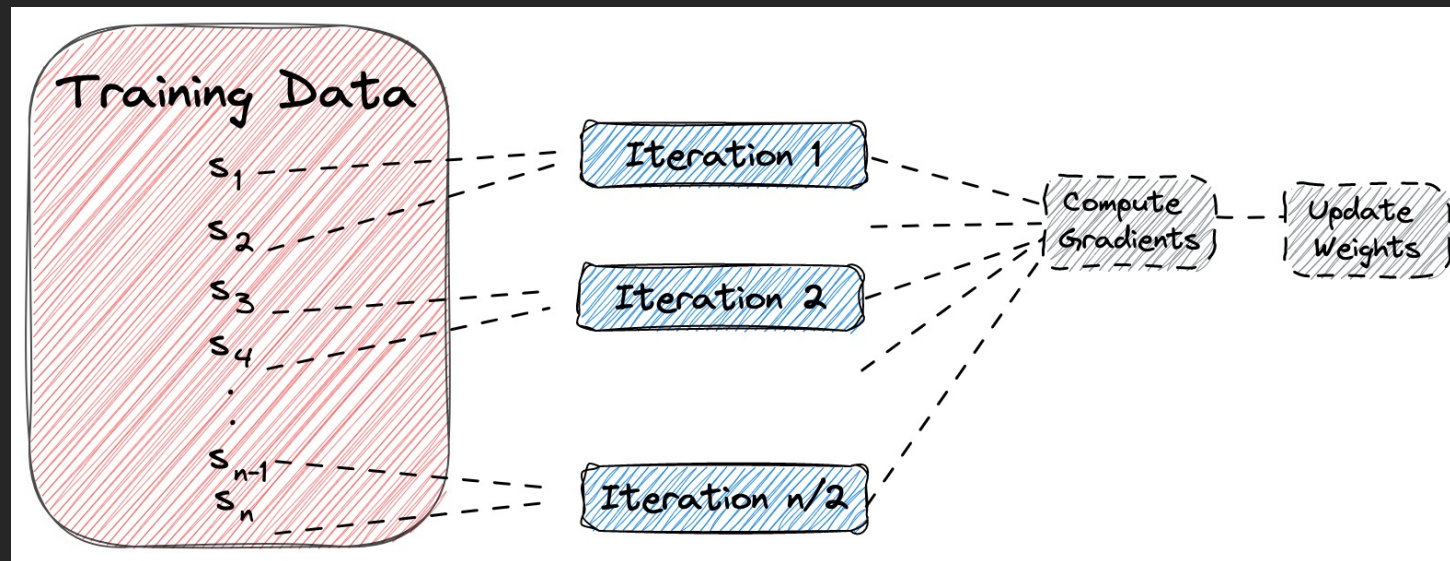
- *choice of the batch size*

Convergence will be less stable depending on the data fluctuations in the batch

- Smaller the batch size, the greater the randomness
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Mini batch learning

- Uses part of the data
- Computes an estimate of the gradient

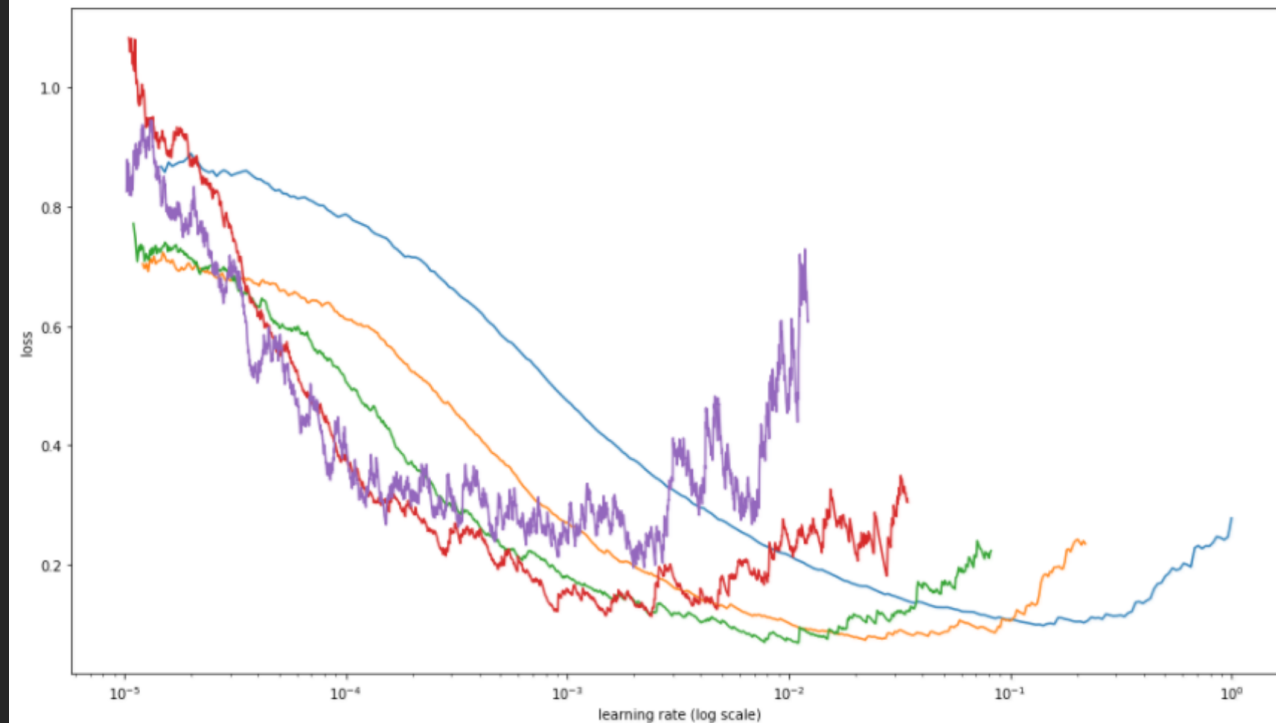


Trade-offs

- Full batch is rarely used in production since it's only suitable for very small datasets
- The choice of a batch size is an important decision

LOSS vs. LEARNING RATE FOR DIFFERENT BATCH SIZES

BATCH SIZE:



Source: <https://miguel-data-sc.github.io/2017-11-05-first/>

The extreme case... streaming data

- Multiple passes on data are no longer possible.

Data needs to be processed in one pass.

- Data streams may evolve over time

Model can become outdated.

The high speed needed to stream data

Online learning

- Model parameters are updated whenever a new observation arrives
- The model is always adjusting to changes in the data

How fast should the model adapt?

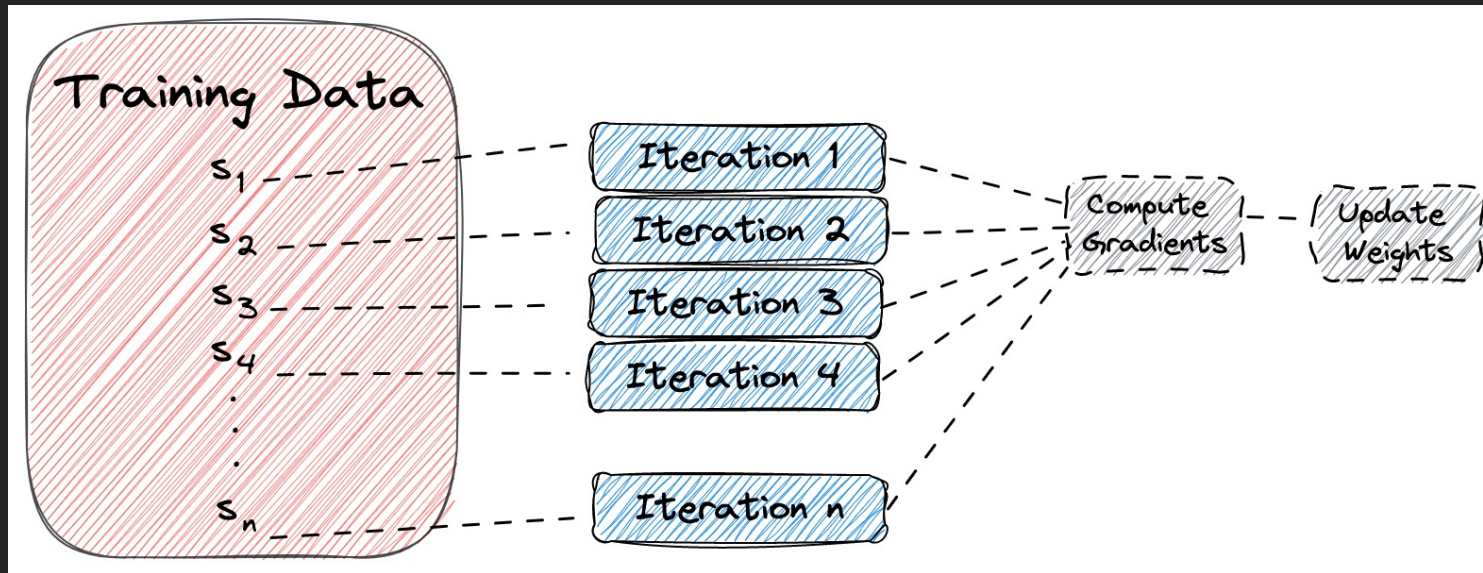
- Bad data will have immediate performance impact
- Controlled by the learning rate

Will require a closer monitoring

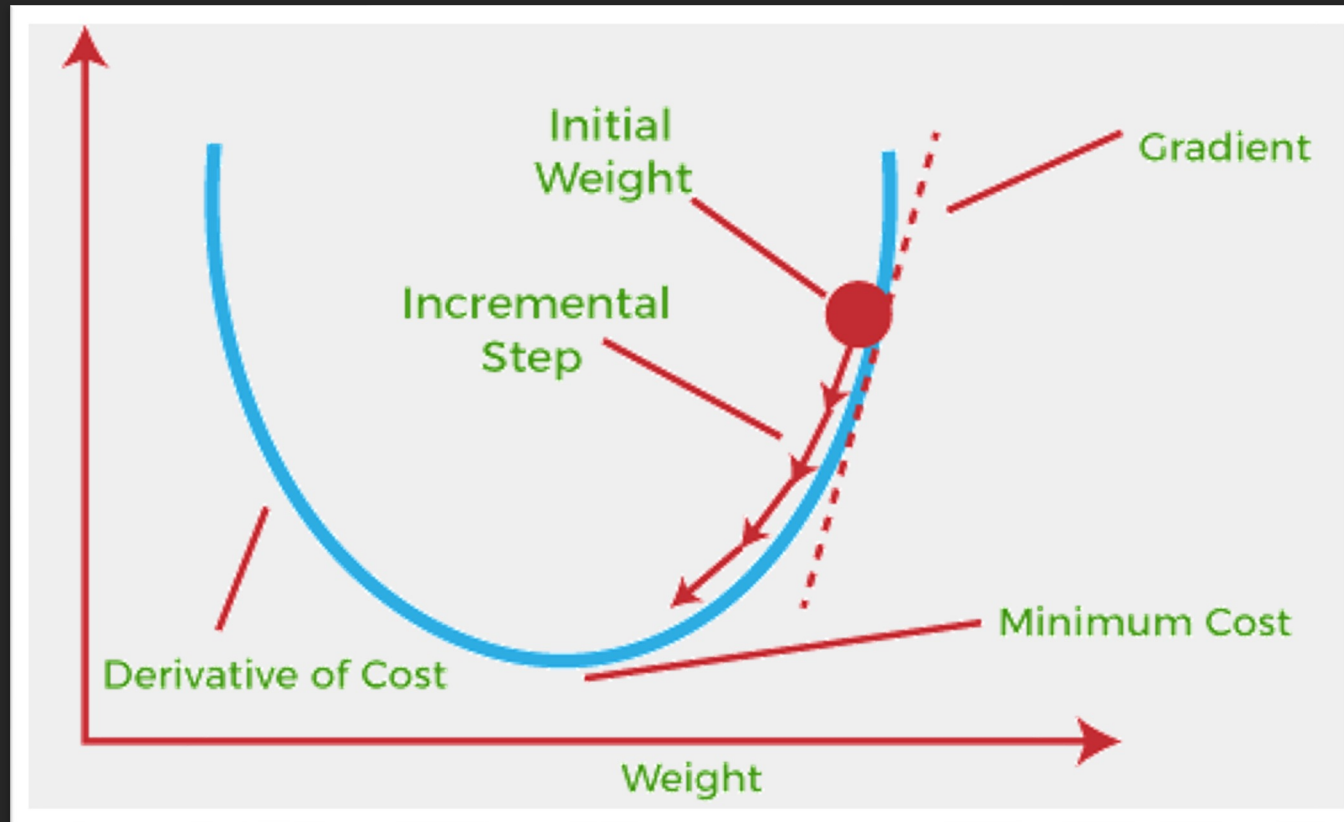
Convergence will be less stable

Online learning

- Uses one single data point per update
- Stochastic gradient descent



Gradient descent



Online learning

- There are three types of data shifts that will cause model decay
 - Covariate shifts
Changes in the input variables from the training data to the test data while the target variables are unchanged
 - Face recognition not being able to identify people wearing masks
 - Prior probability shifts
Changes in the target variables while input variables are unchanged
 - A model that could estimate the probability of someone having flu based on their symptoms during the COVID pandemic
 - Concept drifts
The relationship between input and output change over time in unforeseen ways
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